



2 Real Analysis 2023

Tutorial 6

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Series

Problem 21 Prove or disprove:

- (a) If $\sum_{n=1}^{\infty} a_n$ converges and $\sum_{n=1}^{\infty} b_n$ diverges, then $\sum_{n=1}^{\infty} (a_n + b_n)$ diverges.
- (b) Assume that $a_n > 0$ for all $n \in \mathbb{N}$. If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} \frac{1}{a_n}$ diverges.
- (c) Assume that $a_n > 0$ for all $n \in \mathbb{N}$. If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} \frac{1}{a_n}$ converges.

Problem 22 For each of the following series, determine whether it converges or diverges.

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|---|---|---|
| (a) $\sum_{n=3}^{\infty} \frac{n+4}{n^2-3n+1},$ | (b) $\sum_{n=0}^{\infty} \frac{\sqrt{n+1}}{n^2+1},$ | (c) $\sum_{n=1}^{\infty} \frac{3+\sin n}{n^3},$ |
| (d) $\sum_{n=0}^{\infty} \sqrt{\frac{n+1}{n^2+1}},$ | (e) $\sum_{n=1}^{\infty} \frac{1+2n}{3n},$ | (f) $\sum_{n=1}^{\infty} \frac{1}{n(2n-5)},$ |
| (g) $\sum_{n=1}^{\infty} \frac{1}{n2^n},$ | (h) $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^n,$ | (i) $\sum_{n=2}^{\infty} \frac{\ln n}{n^3}.$ |

Subsequences

Problem 23

- (a) Describe all convergent subsequences of the sequence

$$1, -1, 1, -1, 1, -1, \dots$$

Hint: There are infinitely many convergent subsequences, but they can have only two possible limits.

- (b) Give an example of a sequence that has convergent subsequences with three different limits.

Problem 24 Prove the following statement:

Let $(a_n)_{n=1}^{\infty}$ be a sequence. If $\lim_{k \rightarrow \infty} a_{2k} = \lim_{k \rightarrow \infty} a_{2k-1} = L$, then $\lim_{n \rightarrow \infty} a_n = L$.